



Supermarket Sales & Performance Analysis

Data Analytics Portfolio
Project

EXECUTIVE SUMMARY

Category	Key Metrics
Top Performing Branch	Giza leads in total sales (110,568.71 MMK) and customer satisfaction score (7.07 / 10).
Lowest Rated Branch	Cairo tied Alex in sales (~106.20K MMK) but recorded the lowest rating (6.82 / 10).
Top Product Line	Food and beverages generates the highest sales (56,144.84 MMK) and gross income (2,673.56 MMK).
Lowest Product Line	Health and beauty trails all categories in gross income (2,342.56 MMK).
Monthly Trend	Peak sales occurred in January (116,291.87 MMK), dropped sharply in February (97,219.37 MMK), and rebounded in March (109,455.51 MMK).

Demographics & Loyalty	Members generate 42.3% more sales (189,694.76 MMK) than non-members. Female buyers account for ~60.3% of total revenue (194,671.84 MMK).
Payment Preferences	Cash is the top payment method (112,206.57 MMK), followed closely by E-Wallet (109,993.11 MMK) and Credit Card (100,767.07 MMK).

BUSINESS PROBLEM

The supermarket management team wants to better understand the company's sales and profitability performance across its branches, product lines, and customer segments. Management is concerned that some branches and products may be underperforming and wants to identify which areas are contributing most to revenue and gross income. They also want to understand customer purchasing behavior and satisfaction to support better business decisions.

Goal: Examine historical sales data, identify key performance trends and areas for improvement, and provide data-driven recommendations that can help management improve sales performance and profitability.

BUSINESS QUESTIONS

- Which branch performs best and which needs improvement based on sales?
- Which product line generates the highest sales and gross income?
- Do Member or Normal customers generate more sales?
- How do sales and gross income change over time?
- Which payment method is most commonly used?
- Which branch has the highest customer satisfaction?
- How do sales differ across genders?



DATA QUALITY CHECK

Reviewing the dataset before analysis

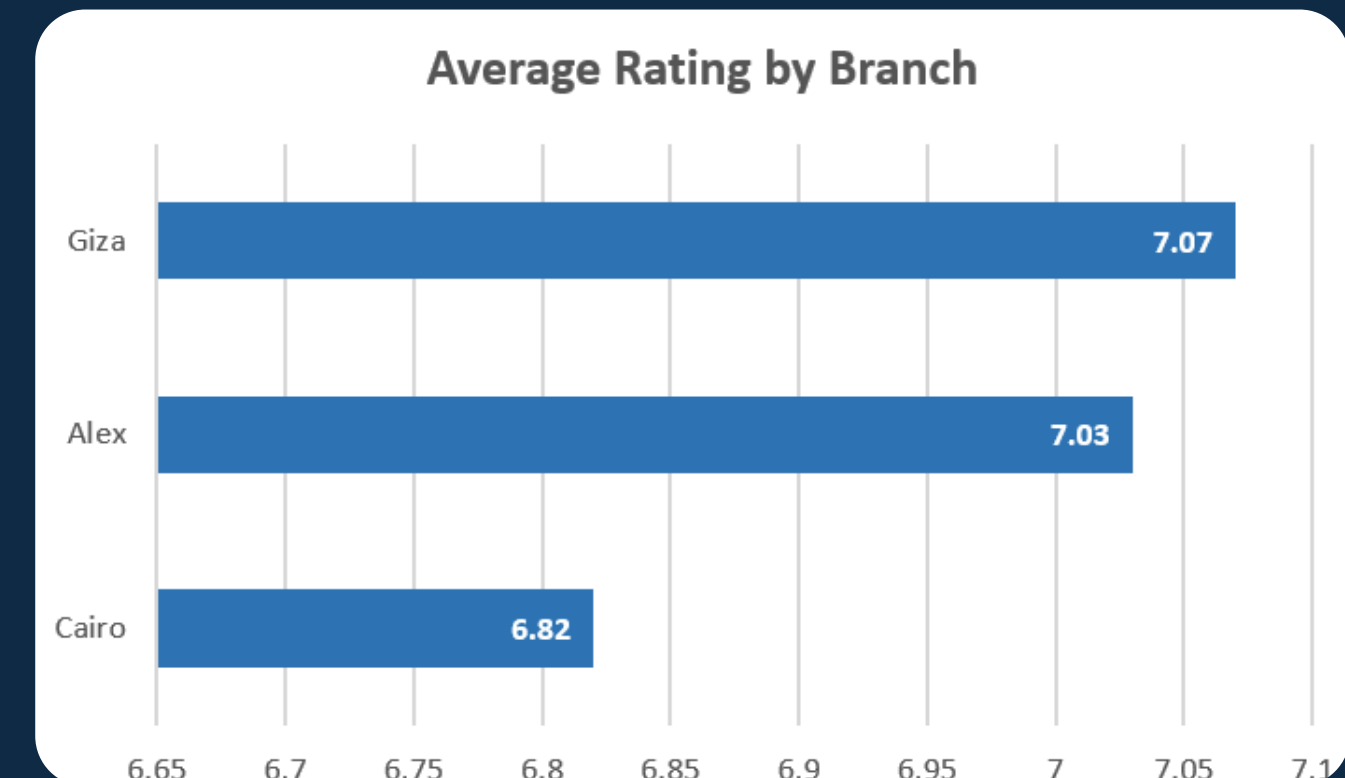
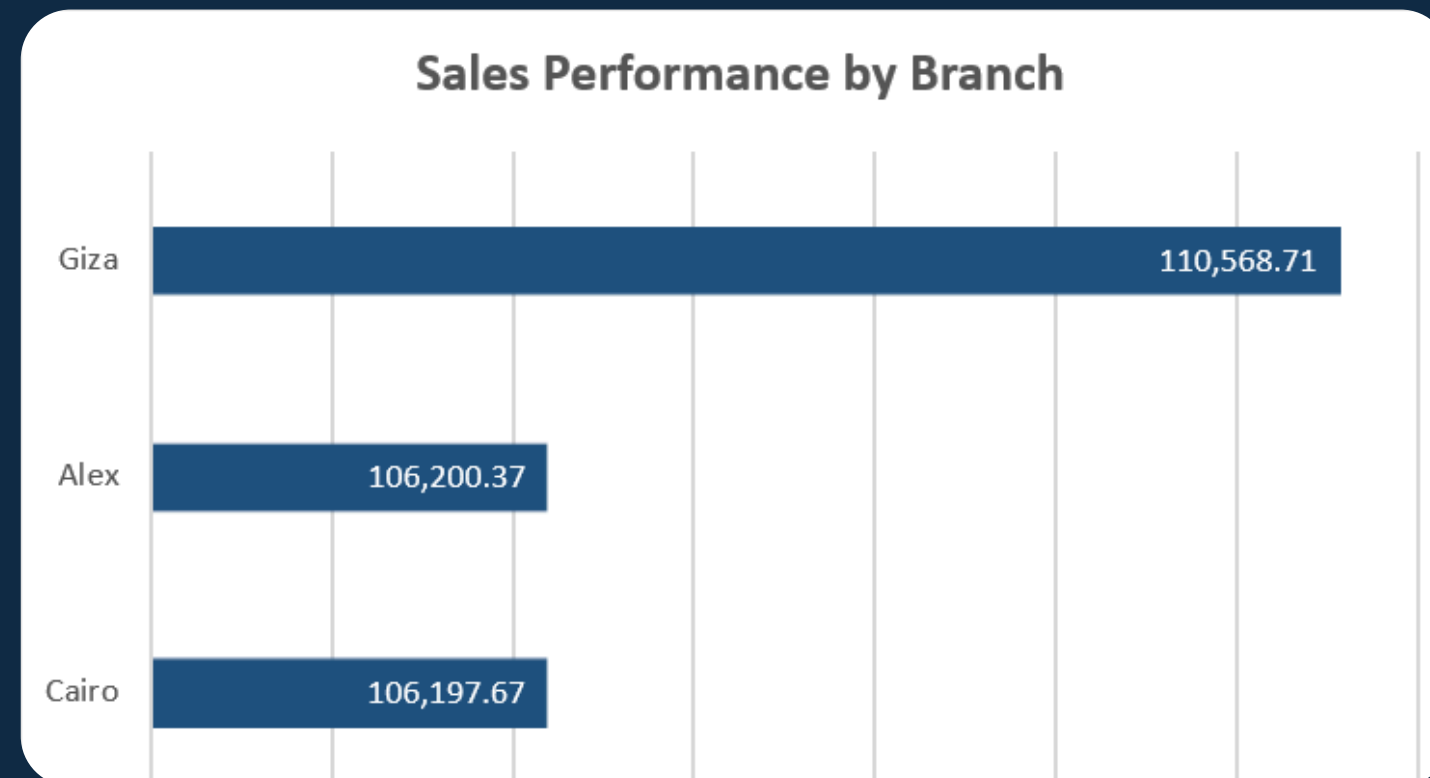
Check	Result	Action
Duplicate Records	No major issues found	No removal required
Missing Values	No major issues found	No imputation required
Data Types	Appropriate for analysis	No major changes required
Category Values	Consistent	No standardization required
Numerical Fields	Valid and Usable	Ready for analysis
Date Field	Valid	Ready for time analysis

Conclusion: The dataset was already sufficiently clean and structured, so no major data cleaning was required.

FINDINGS

Currency in MMK (Myanmar Kyat)

Which branch generated the highest sales volume and customer satisfaction?

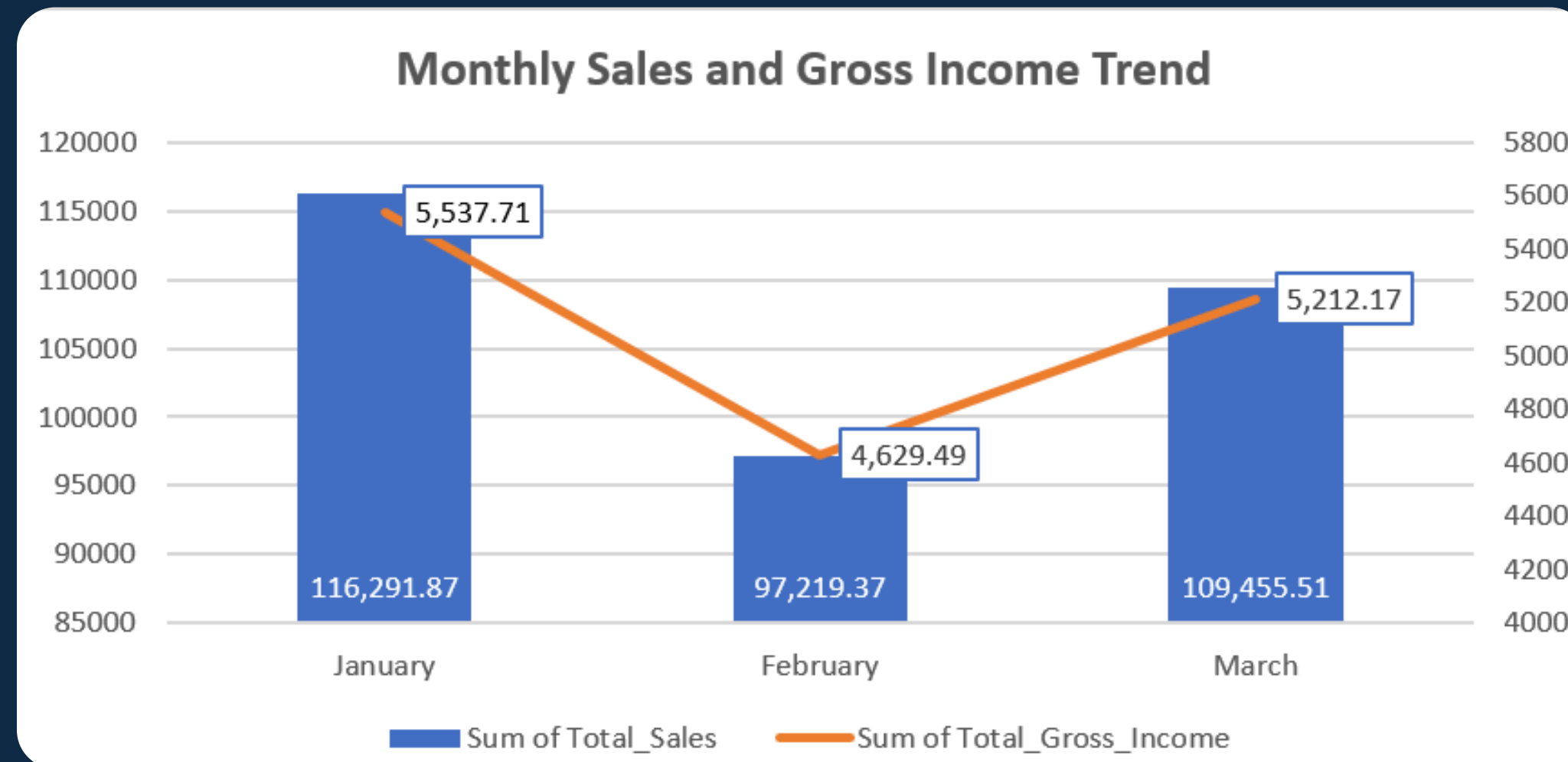


Giza leads overall performance, driving **110,568.71 MMK** in revenue with a **7.07/10** satisfaction score. Cairo tied Alex in revenue (~106.20K MMK) but recorded the lowest customer rating (6.82/10).

FINDINGS

Currency in MMK (Myanmar Kyat)

How do sales and gross income change over time?



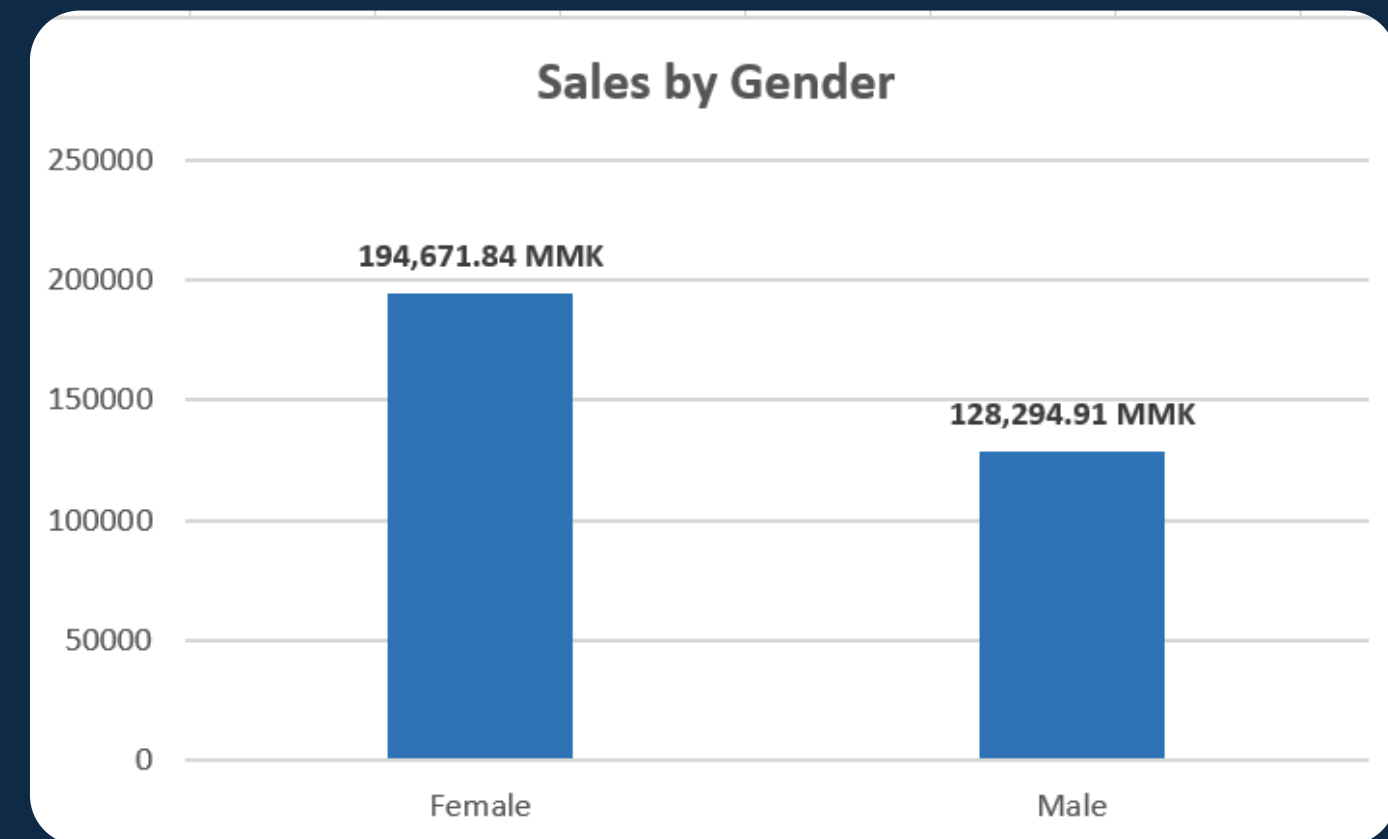
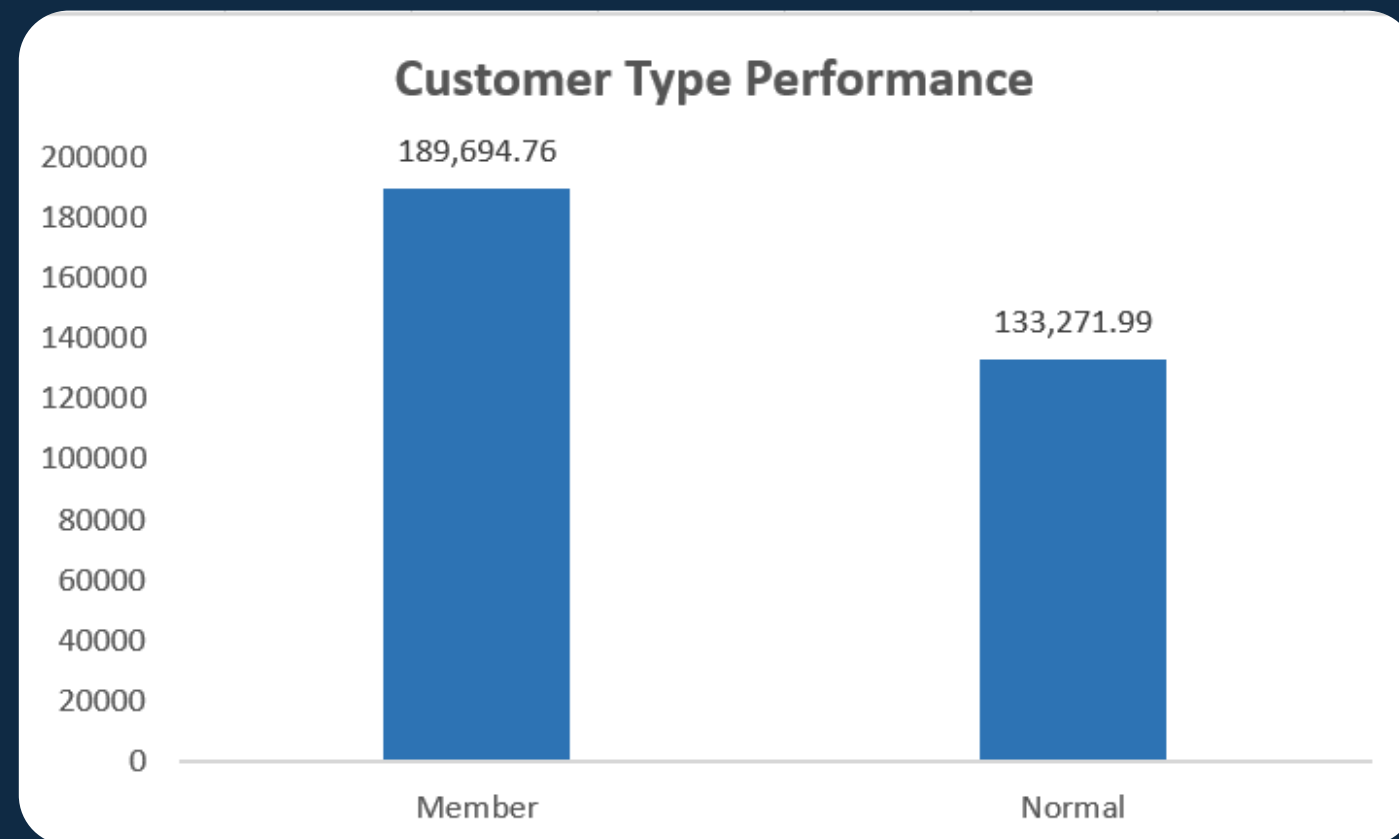
January achieved the highest performance across the *3-month* period with **116,291.87 MMK** in sales and **5,537.71 MMK** in gross income.

Revenue experienced a sharp drop in February (**97,219.37 MMK** sales / **4,629.49 MMK** gross income) before rebounding in March (109,455.51 MMK sales / 5,212.17 MMK gross income).

FINDINGS

Currency in MMK (Myanmar Kyat)

Which Customer Segments and Demographics Drive the Highest Revenue?



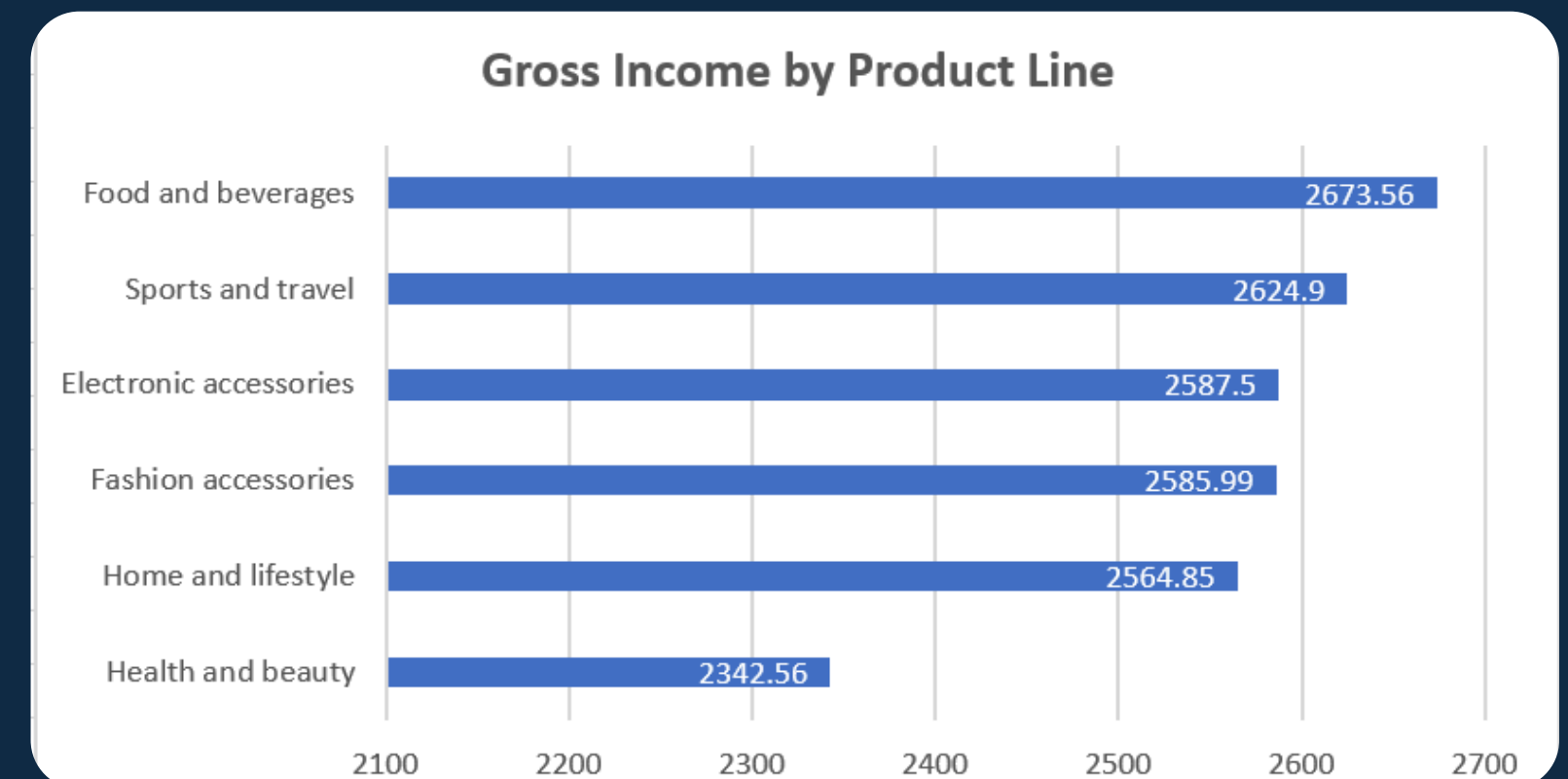
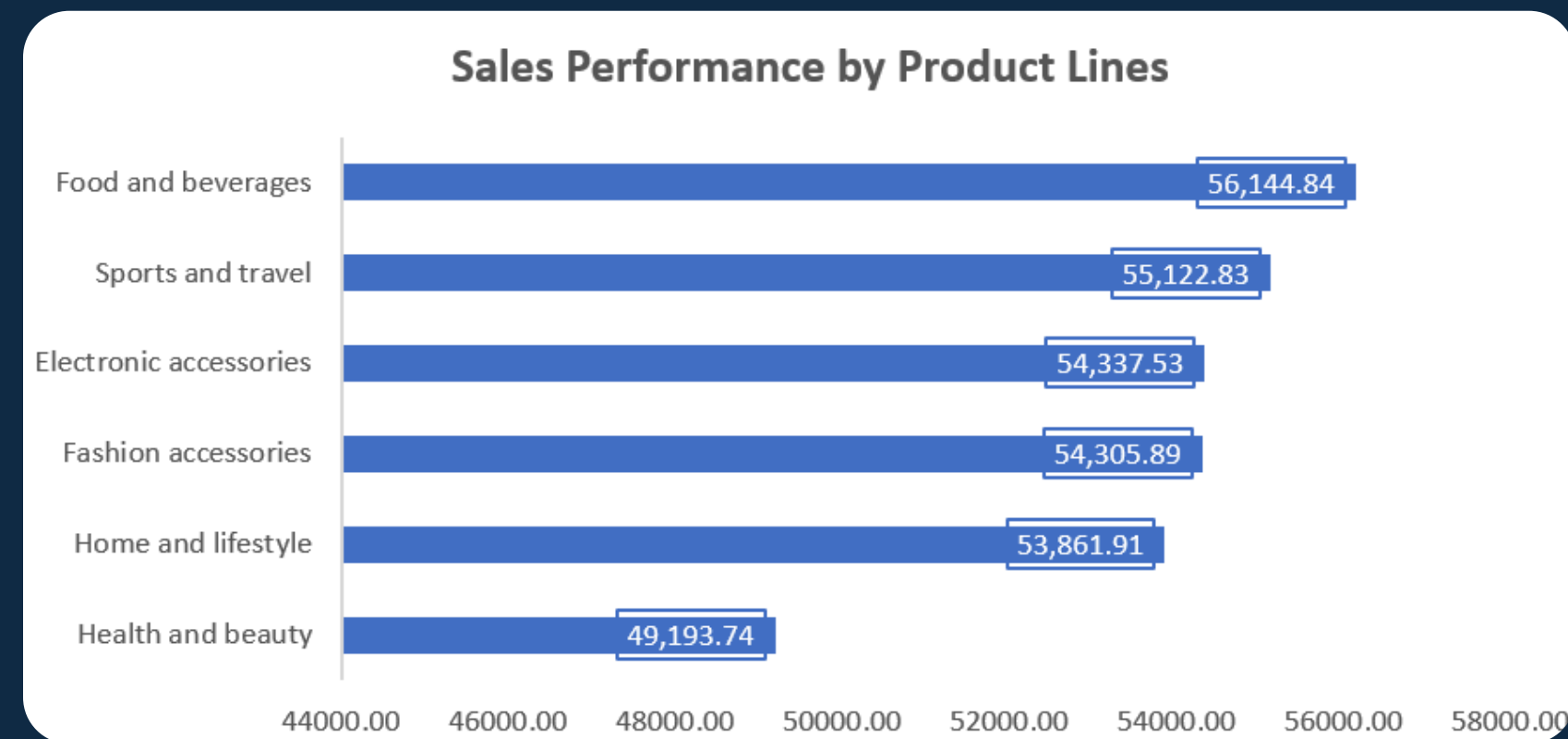
Members generate 42.3% more sales (**189,694.76 MMK**) than non-members/normal customers (**133,271.99 MMK**), underlining the value of loyalty programs.

Female buyers account for significantly higher sales (**194,671.84 MMK**) compared to Male buyers (**128,294.91 MMK**).

FINDINGS

Currency in MMK (Myanmar Kyat)

Which Product Lines Drive the Highest Revenue and Gross Profit?



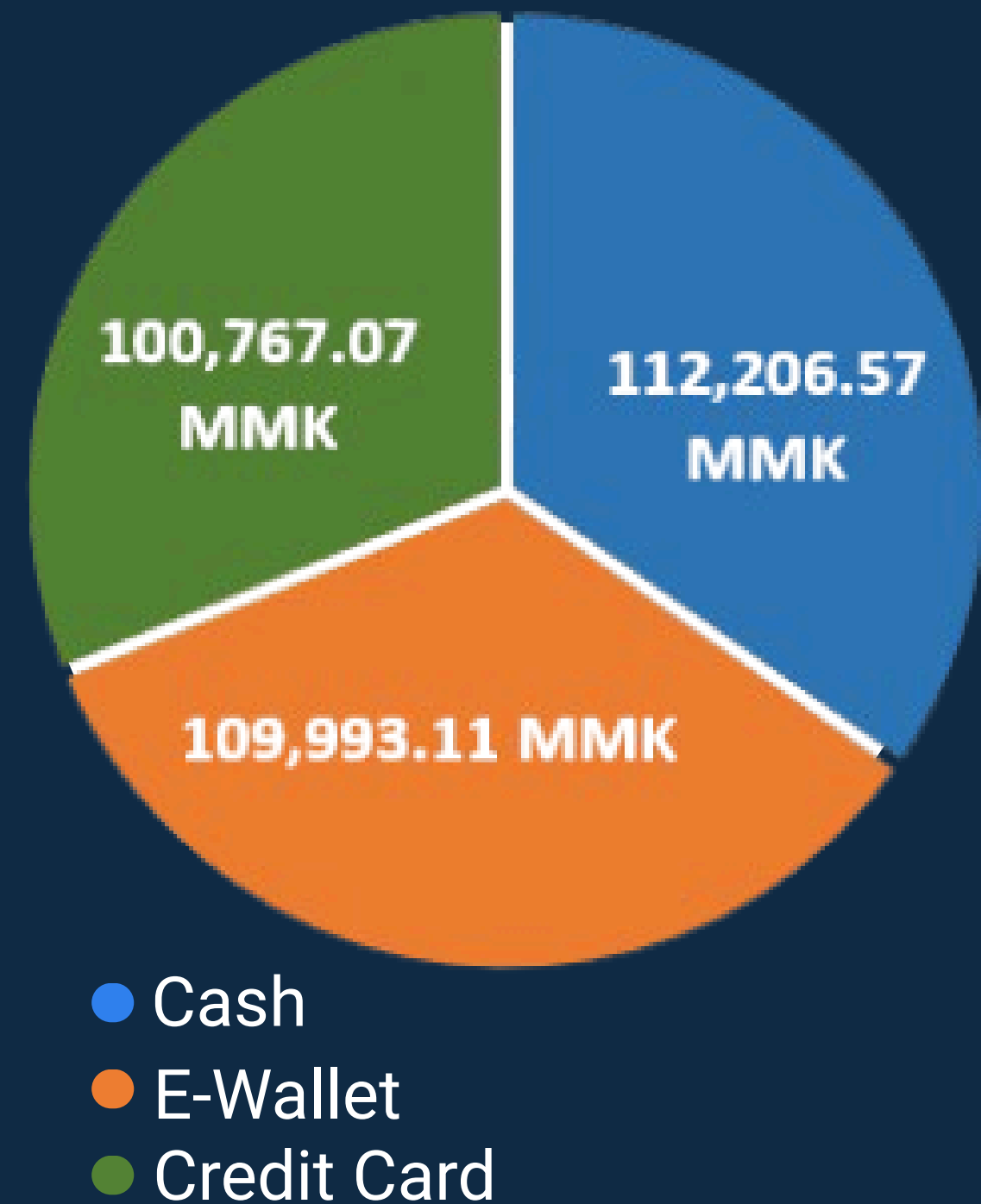
Food and beverages leads gross profit at **2,673.56 MMK**, followed closely by Sports and travel (**2,624.90 MMK**). Health and beauty trails all categories substantially at 2,342.56 MMK in gross income.

FINDINGS

Currency in MMK (Myanmar Kyat)

Which payment method is most commonly used?

Cash is the dominant transaction channel (**112,206.57 MMK**), followed closely by E-Wallet (109,993.11 MMK) and Credit Card (100,767.07 MMK).



KEY INSIGHTS

Members generate 42.3% more sales (**189,694.76 MMK**) than non-members (133,271.99 MMK), proving that repeat customers are the core revenue engine.

Cash are the top payment channel (112,206.57 MMK), and female shoppers account for 60.3% of total revenue (194,671.84 MMK vs. 128,294.91 MMK for males).

Monthly revenue dropped sharpest in February (97,219.37 MMK) compared to January's peak (116,291.87 MMK), indicating *seasonal demand slumps or mid-quarter inventory bottlenecks*.

RECOMMENDATIONS

Turn Guest Buyers into Members

Why: Members spend significantly more money than non-members (189.7K MMK vs. 133.3K MMK).

Action: Offer cashiers a small bonus or give customers a discount on their first purchase when they sign up for a loyalty membership at checkout.

Fix Service Issues at the Cairo Branch

Why: Cairo makes as much money as other stores (~106.2K MMK), but customers leave lower ratings (6.82 out of 10).

Action: Retrain Cairo staff on customer service and shorten checkout wait times so happy customers keep coming back.

RECOMMENDATIONS

Boost Sales During Slow Months (February)

Why: Sales dropped in February (97.2K MMK) compared to January (116.3K MMK).

Action: Run special discounts or promotions in mid-February—especially on popular items like Food & Beverages—to keep store sales steady.

Promote Digital Payments and Cater to Female Shoppers

Why: Cash is still the primary payment method (112.2K MMK), but E-wallets are a close second (110.0K MMK). Female shoppers also generate the majority of sales (194.7K MMK).

Action: Encourage cash users to adopt E-wallets to speed up checkout queues, and stock more inventory aimed at female buyers.

Conclusion / Next Steps

Summary of Where We Are

- Our stores are healthy and bringing in steady money.
- Most of our income comes from female shoppers, loyalty members, and cash/E-wallet transactions.
- Stores are selling well, but Cairo needs better customer service, and February needs a sales boost.

What Management Should Do First

Talk to the Cairo store manager to figure out why customer ratings are lower than other branches.

Set up a simple loyalty program pitch at registers to convert normal shoppers into members.

Plan a February promotional calendar so sales don't drop again next year.



THANK YOU

Questions & Discussion

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TECHNICAL APPENDIX

Click here:

<https://drive.google.com/file/d/1i96lZseiUs54en6ZO-lx06HOMtNQQ6O6/view?usp=sharing>